

- ❖ **Littoral zone:** It is the shallow part of the lake closer to the shore. Light penetrates up to the bottom. It is 'euphotic' (having good light), has rich vegetation and higher rate of photosynthesis, hence rich in oxygen.
- ❖ **Limnetic zone:** It is the open water zone away from the shore. It extends up to the **effective light penetration level**, vertically. The imaginary line that separates the limnetic zone from the profundal zone is known as **zone of compensation/ compensation point/ light compensation level**. It is the zone of effective light penetration. Here the rate of photosynthesis is equal to the rate of respiration. Limnetic zone has no contact with the bottom of the lake.
- ❖ **Profundal zone:** It is the deep water area present below the limnetic zone and beyond the depth of effective light penetration. Light is absent. Photosynthetic organisms are absent and so the water is poor in oxygen content. It includes mostly the anaerobic organisms which feed on detritus.

The organisms living in lentic habitat are classified into **pedonic** forms, which live at the bottom of the lake and those living in the open waters of lakes, away from the shore vegetation are known as **limnetic forms**.

Biota (animal and plant life of a particular region) of the littoral zone

Littoral zone is rich with pedonic flora (especially up to the depth of the effective light penetration). At the shore proper '**emergent vegetation**' is abundant with firmly fixed roots in the bottom of the lake and shoots and leaves are exposed above the level of water. These are **amphibious plants**. Certain emergent rooted plants of littoral zone are the cattails (*Typha*), bulrushes (*Scirpus*), arrowheads (*Sagittaria*). Slightly deeper are the rooted plants with floating leaves, such as the water lilies (*Nymphaea*), *Nelumbo*, *Trapa*, etc. Still deeper are the submerged plants such as *Hydrilla*, *Chara*, *Potamogeton*, etc., The free floating vegetation includes *Pistia*, *Wolffia*, *Lemna* (duckweed), *Azolla*, *Eichhornia*, etc.

The **phytoplankton** of the littoral zone composed of diatoms (*Coscinodiscus*, *Nitzschia*, etc.), green algae (*Volvox*, *Spirogyra*, etc.), euglenoids (*Euglena*, *Phacus*, etc.), and dinoflagellates (*Gymnodinium*, *Cystodinium*, etc.).

Animals, the **consumers** of the littoral zone, are abundant in this zone of the lake. These are categorized into zooplankton, neuston, nekton, periphyton, and benthos. The **zooplankton** of the littoral zone consists of 'water fleas' such as *Daphnia*, rotifers and ostracods.

The animals living at the *air-water interface* constitute the '**neuston**'. They are of two types, the epineuston and hyponeuston. Water striders (*Gerris*), beetles, water bugs (*Dineutes*) form the **epineuston/supraneuston** and the **hyponeuston/infraneuston** includes the 'larvae of mosquitoes'.

The animals such as fishes, amphibians, water snakes, terrapins, insects like 'water scorpion' (*Ranatra*), 'back swimmer' (*Notonecta*), 'diving beetles' (*Dytiscus*), capable of swimming constitute the **nekton**.

The animals that are attached to / creeping on the aquatic plants, such as the 'water snails', 'nymphs of insects', 'bryozoans', 'turbellarians', 'hydras', etc., constitute the '**periphyton**'.

The animals that rest on or move on the bottom of the lake constitute the '**benthos**', e.g., red annelids, chironomid larvae, cray fishes, some isopods, amphipods, clams, etc.

Biota of the limnetic zone

Limnetic zone is the largest zone of a lake. It is the region of rapid variations of the level of the water, temperature, oxygen availability, etc., from time to time. The limnetic zone has autotrophs (photosynthetic plants) in abundance. The chief autotrophs of this region are the phytoplankton such as the **euglenoids, diatoms, cyanobacteria, dinoflagellates** and **green algae**. The consumers of the limnetic zone are the zooplanktonic organisms such as the copepods. Fishes, frogs, water snakes, etc., form the limnetic **nekton**.

Biota of the profundal zone

It includes the organisms such as decomposers (bacteria), *chironomid* larvae, *Chaoborus* (**phantom larva**), red annelids, clams, etc., that are capable of living in low oxygen levels. The decomposers of this zone decompose the dead plants and animals and release nutrients which are used by the biotic communities of both littoral and limnetic zones.

The lake ecosystem performs all the functions of any ecosystem and of the biosphere as a whole, i.e., conversion of inorganic substances into organic material, with the help of the radiant solar energy by the autotrophs; consumption of the autotrophs by the heterotrophs; decomposition and mineralization of the dead matter to release them back for reuse by the autotrophs (**recycling of minerals**).

8.4.4 The Terrestrial Ecosystems

The ecosystems of land are known as terrestrial ecosystems. Some examples of terrestrial ecosystems are the forest, grassland and desert.

- i. The Forest Ecosystems:** The two important types of forests seen in India are i) tropical rain forest and ii) tropical deciduous forests.
- ii. The Grassland Ecosystems:** These are present in the Himalayan region in India. They occupy large areas of sandy and saline soils in Western Rajasthan.
- iii. Desert Ecosystems:** The areas having less than 25cm rainfall per year are called deserts. They have characteristic flora and fauna. The deserts can be divided into two types – hot type and cold type deserts. Thar Desert in Rajasthan is the example for hot type of desert. Cold type desert is seen in Ladakh.

8.4.5 Artificial Ecosystems

These are man-made ecosystems such as agricultural or agro-ecosystems. They include cropland ecosystems, aquaculture ponds and aquaria.